

Supplementary Materials – Characterisation of L4.13, a cryptic European sub-lineage of *Mycobacterium tuberculosis* with focal signatures of positive selection on ESX-2/4 secretion and PDIM biosynthesis

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Supplementary tables

The following tables are provided as tab-separated files in the `supplementary_materials/` directory of the project repository. Each is self-describing through its header row.

Table S1 — Strains with metadata. File: `table_S1_strains.tsv`. One row per strain (258 L4.13 + 10 outgroup), columns: strain accession, sub-lineage assignment, SPDI count, QC metrics (missing genes, missing RD, mean depth, covered bases percent, mean MAPQ, mean BASEQ, GC content), and ENA metadata (country, collection date, host, BioProject, BioSample, platform).

Table S2 — Core-exclusive SPDIs with annotation. File: `table_S2_core_exclusive.tsv`. One row per SPDI (249 total: 187 L4.13-global, 23 L4.13.1-specific, 39 L4.13.2-specific). Columns: SPDI identifier, clade (L4.13 / L4.13.1 / L4.13.2), position, ref/alt, gene name, locus tag, strand, predicted effect (`synonymous_variant`, `missense_variant`, `stop_gained`, `frameshift_variant`, `deletion`, `intergenic_region`), impact severity, amino-acid change where applicable, gene product, and Mycobrowser functional category.

Table S3 — SNP marker validation. File: `table_S3_marker_validation.tsv`. For each of the three proposed markers (L4.13 3222055:G:A, L4.13.1 187910:T:C, L4.13.2 1129237:G:A), rows reporting inclusion penetrance in the target clade (L4.13.1 n=61, L4.13.2 n=197) and exclusion leakage in each of 17 tested sister sub-lineages (up to 50 strains per sister, seed 42, total n=1485).

Table S4 — In silico spoligotypes. File: `table_S4_spoligotypes.tsv`. Per-strain 43-spacer binary pattern (columns) and 15-digit SITVIT2-style octal code, for the 188 strains with complete `known_coverage` data. SITVIT2 returned zero matching isolates for all five most-frequent octals of our dataset (see main text).

Table S5 — Stratified McDonald-Kreitman test. File: `table_S5_mk_stratified.tsv`. Six functional strata (global, PDIM strict, PDIM extended, ESX strict, ESX extended, PE/PPE, rest-of-genome), with D_n , D_s , P_n , P_s , D_n/D_s : P_n/P_s , two-sided and one-sided Fisher exact p-values, Direction of Selection (DoS) with 10 000-bootstrap 95% confidence interval.

Supplementary figures

Supplementary methods

Ancient-sample anchors for LSD2 dating. The IQ-TREE 2.4.0 + LSD2 pipeline (main text, Molecular dating section) used the following nine ancient calibrators retrieved from the SPAAM ancient metagenome directory: three pinnipedii genomes from Bos et al. 2014 (SRR1238557, SRR1238558, SRR1238559, ^{14}C date b(1028,1280)); three pinnipedii genomes from Vågene 2022 (S82 b(1250,1470), S281 b(1265,1380), S386 b(1260,1400)); Winstrup’s Lund 1 (Sabin 2020, historical date 1679);

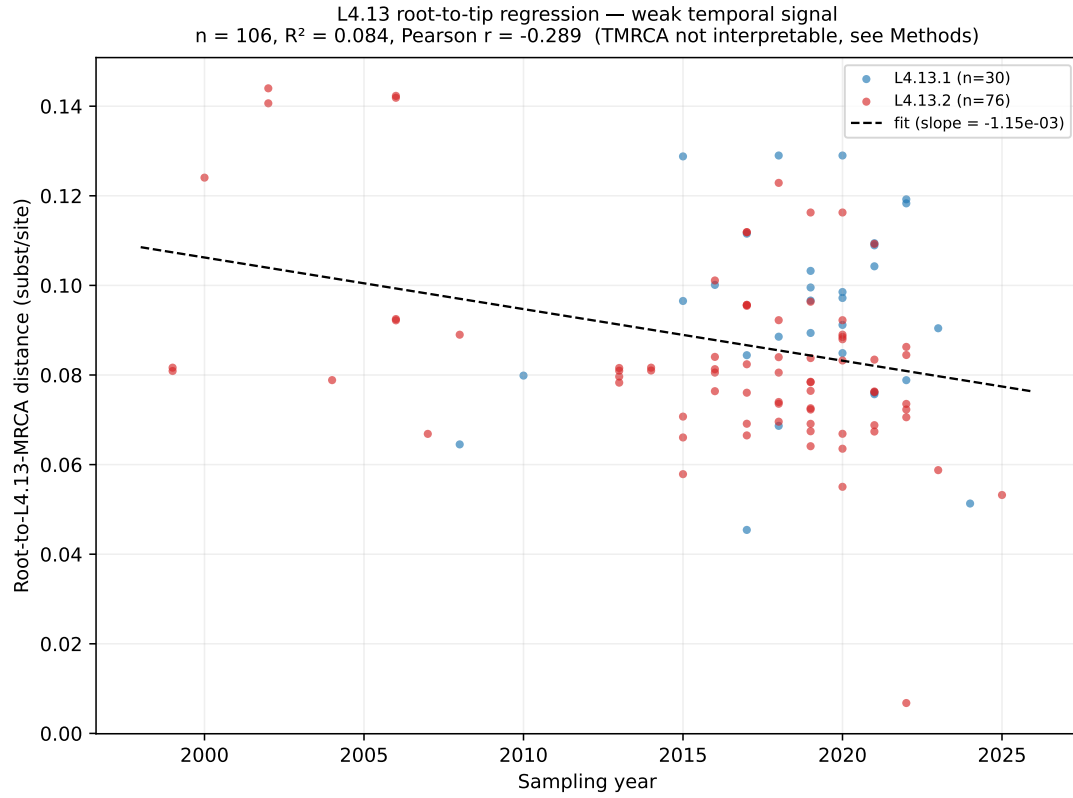


Figure 1: **Supplementary Figure S1 — Root-to-tip regression.** Scatter of root-to-L4.13-MRCA distance (in BIN+G substitutions/site) vs sampling year for the 104 L4.13 tips with known collection date. Blue: L4.13.1 ($n=28$ dated tips). Red: L4.13.2 ($n=76$ dated tips). Linear regression yields $R^2 = 0.09$ and a negative slope, demonstrating that the BIN+G binary tree carries insufficient temporal signal for reliable tip-date-only molecular clock inference. This motivates the ancient-sample-anchored LSD2 approach used in the main text.

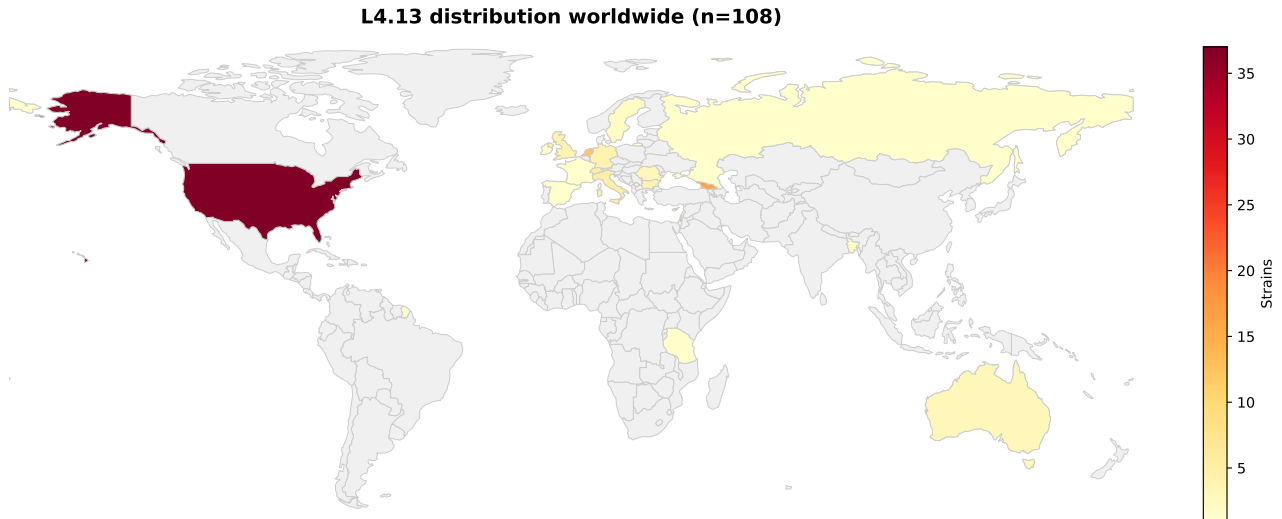


Figure 2: **Supplementary Figure S2 — Global choropleth distribution of L4.13.** Colour scale (YlOrRd, log-transformed) encodes the number of L4.13 strains per country among the 108 strains with ENA-declared country of origin. Countries in grey are either absent from our sample or have **Unknown** / **not provided** metadata.

and two Kay 2015 Vác body samples (Body 68 cleaned for mixed infection via inter-L4 sharing, Body 92, $b(1731,1838)$ historical interval). The H37Rv MRCA was constrained to $b(1900,1910)$ as

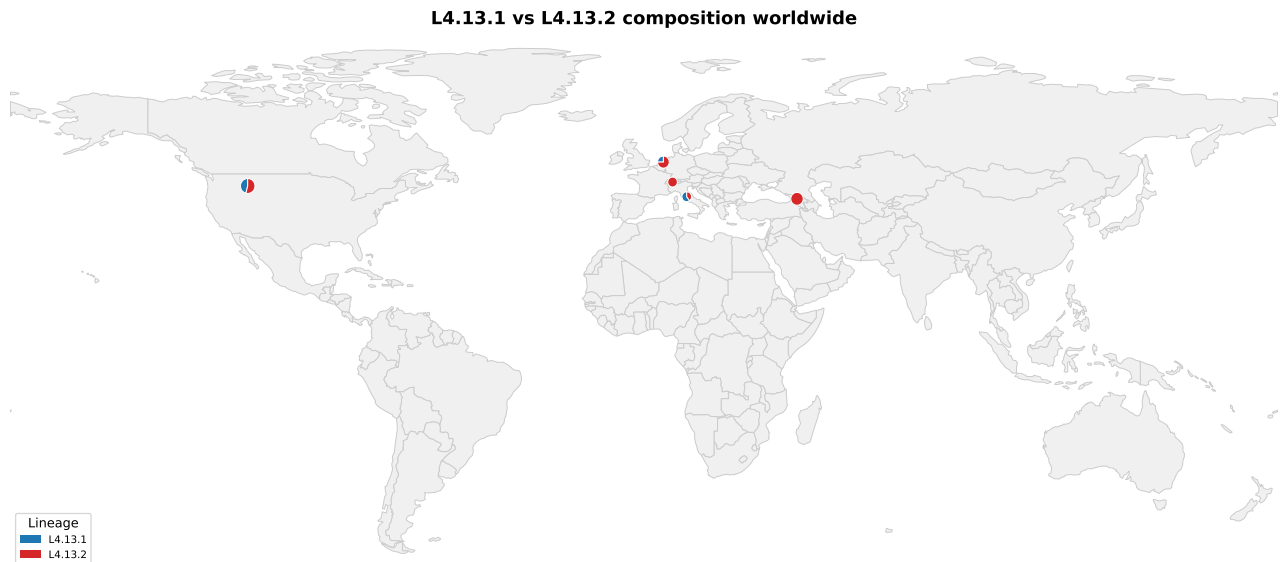


Figure 3: **Supplementary Figure S3 — Global sub-clade composition.** Pie charts on each country with $n \geq 3$ showing the proportion of L4.13.1 (blue) vs L4.13.2 (red). The United States is the only region with substantial co-circulation of both sub-clades; Georgia is exclusively L4.13.2.

an additional nodal anchor. The multi-seed workflow (5 seeds, objective function selection) selected seed 335004 as the best run (LSD2 objective = 0.432), yielding the point estimates quoted in the main text.

Binary alignment and outgroup rationale. The 268-taxon binary alignment used 5 049 parsimony-informative sites (SPDIs present in ≥ 2 and absent in ≥ 2 strains). The outgroup was chosen on the basis of the global L4 topology (`lignees.py[moi]`) which places L4.13 as sister to the L4.4-containing clade: 5 L4.4.2 strains (direct sister) plus 5 L4.2.1 strains (distal L4 anchor) were selected at random with seed 42.

Reproducibility. All analysis scripts (phases 1 through 9) are archived in the `analyses/` directory of the project repository. Each phase script is self-contained and reproducibly executes its corresponding analysis from the raw TBannotator SPDI profiles and ENA metadata.